



Inventory and Branches Checklist

This checklist covers the Inventory application and its branch system inside Orviohub.

The structure is:

```
User
├── Organization
│   ├── Inventory application
│       ├── Branches
│           └── Products, stock, sales, purchases, and staff
```

Inventory must remain organization-scoped, application-scoped, and branch-scoped. The PRD requires every stock change to create a stock movement, posted business records to remain historically traceable, and sale confirmation plus stock deduction to be atomic. ^[1]

1. Inventory Application Foundation

Application registration

- ☐ Inventory exists in the application registry.
- ☐ Inventory is marked available.
- ☐ Inventory is visible to regular users.
- ☐ Inventory is activatable.
- ☐ Future applications remain hidden.
- ☐ Inventory has a unique application key.
- ☐ Inventory has a display name.
- ☐ Inventory has a description.
- ☐ Inventory has an icon.
- ☐ Inventory has a route or subdomain.
- ☐ Inventory status is controlled server-side.
- ☐ Application status changes are audited.
- ☐ Inventory activation is separate from user registration.
- ☐ Inventory activation is separate from organization creation.
- ☐ Inventory activation requires an eligible organization.

Inventory access

- ☐ User is authenticated.
- ☐ User account is active.
- ☐ User has active organization membership.
- ☐ Organization is not deleted.
- ☐ Organization is not suspended unless limited access is allowed.
- ☐ Inventory is enabled for the organization.
- ☐ User has Inventory membership.
- ☐ User has an Inventory role.
- ☐ User has branch access where required.
- ☐ Billing entitlement permits Inventory access.
- ☐ Branch status permits access.
- ☐ Access is checked server-side.
- ☐ Direct URL manipulation cannot bypass access checks.
- ☐ Client-submitted application keys are validated.
- ☐ Cross-organization access is rejected.
- ☐ Future applications cannot be accessed through Inventory routes.

Inventory application lifecycle

- ☐ `inactive` status exists.
- ☐ `activating` status exists.
- ☐ `setup_incomplete` status exists.
- ☐ `active` status exists.
- ☐ `suspended` status exists.
- ☐ `deactivated` status exists.
- ☐ `expired` status exists.
- ☐ Valid status transitions are defined.
- ☐ Invalid transitions are rejected.
- ☐ Activation is idempotent.
- ☐ Suspension is audited.
- ☐ Restoration is audited.
- ☐ Application errors are visible to authorized administrators.
- ☐ Failed provisioning can be retried safely.

2. Inventory Activation

- ☐ User can open Inventory activation.
- ☐ Backend verifies organization membership.
- ☐ Backend verifies permission.
- ☐ Backend verifies organization status.
- ☐ Backend verifies billing status.
- ☐ Backend verifies Inventory entitlement.
- ☐ Backend verifies application availability.
- ☐ Backend checks application limit.
- ☐ Backend creates Inventory application record.
- ☐ Backend creates Inventory onboarding flow.
- ☐ Backend creates the primary branch.
- ☐ Backend creates default Inventory roles.
- ☐ Backend creates default Inventory settings.
- ☐ Backend writes an audit event.
- ☐ Backend sends an activation notification.
- ☐ User is redirected to Inventory onboarding.
- ☐ Duplicate activation does not create duplicate records.
- ☐ Activation failure is recoverable.
- ☐ Existing organization data is preserved during retry.

Activation user experience

- ☐ Inventory is shown as available.
- ☐ Activation status is visible.
- ☐ User sees current plan.
- ☐ User sees Inventory limits.
- ☐ User sees trial status if applicable.
- ☐ User sees the required setup steps.
- ☐ User can resume incomplete setup.
- ☐ User receives a clear message when billing blocks activation.
- ☐ User receives a clear message when the organization is suspended.
- ☐ User does not see unfinished future applications.

3. Inventory Onboarding

Onboarding flow

- ☐ Inventory welcome screen exists.
- ☐ Business category step exists.
- ☐ Business profile step exists.
- ☐ Primary branch setup step exists.
- ☐ Currency and pricing step exists.
- ☐ Inventory policies step exists.
- ☐ Product setup step exists.
- ☐ CSV import step exists or is planned.
- ☐ Opening stock step exists.
- ☐ Staff invitation step exists.
- ☐ Receipt settings step exists.
- ☐ First-sale tutorial exists.
- ☐ Completion screen exists.
- ☐ Progress indicator exists.
- ☐ User can resume onboarding.
- ☐ User can skip permitted optional steps.
- ☐ Required steps cannot be skipped.
- ☐ Failed steps display useful errors.
- ☐ Failed steps can be retried.
- ☐ Onboarding changes are audited.
- ☐ Onboarding version is stored.
- ☐ Onboarding completion date is stored.

Onboarding data

- ☐ Flow ID.
- ☐ User ID.
- ☐ Organization ID.
- ☐ Product key.
- ☐ Flow version.
- ☐ Current step.
- ☐ Completed steps.
- ☐ Skipped steps.

- ☐ Failed steps.
- ☐ Flow status.
- ☐ Started date.
- ☐ Last updated date.
- ☐ Completion date.

Required onboarding outcomes

- ☐ Inventory application is active or setup-incomplete.
- ☐ Primary branch exists.
- ☐ Branch settings exist.
- ☐ Inventory settings exist.
- ☐ Default roles exist.
- ☐ Product setup state is recorded.
- ☐ Opening stock state is recorded.
- ☐ Staff invitation state is recorded.
- ☐ Receipt configuration state is recorded.
- ☐ User can reach the Inventory dashboard.
- ☐ User can return to incomplete setup later.

4. Inventory Application Settings

General settings

- ☐ Inventory settings page exists.
- ☐ Current organization is displayed.
- ☐ Current Inventory application is displayed.
- ☐ Active branch is displayed.
- ☐ User's Inventory role is displayed.
- ☐ Application status is displayed.
- ☐ Application onboarding status is displayed.
- ☐ Application audit link exists.

Product settings

- ☐ Product naming behavior is configurable.
- ☐ SKU behavior is configurable.
- ☐ Barcode behavior is configurable.

- ☐ Duplicate SKU validation exists.
- ☐ Duplicate barcode validation exists.
- ☐ Categories can be managed.
- ☐ Units can be managed.
- ☐ Product variants are supported or explicitly deferred.
- ☐ Product image support is available or explicitly deferred.
- ☐ Products can be archived.
- ☐ Archived products remain available in history.
- ☐ Archived products do not appear in normal sales search.
- ☐ Product imports are permission-controlled.
- ☐ Product exports are permission-controlled.
- ☐ Product changes are audited.

Stock settings

- ☐ Negative-stock policy exists.
- ☐ Low-stock threshold exists.
- ☐ Stock adjustment policy exists.
- ☐ Stock-count policy exists.
- ☐ Opening stock policy exists.
- ☐ Stock transfer policy exists.
- ☐ Stock movement history is preserved.
- ☐ Stock changes require a movement record.
- ☐ Direct stock balance editing is prohibited.
- ☐ Adjustment reasons are required.
- ☐ Large discrepancies can require approval.
- ☐ Stock settings are organization/application scoped.
- ☐ Branch-specific stock rules are supported where needed.

Sales settings

- ☐ Cash payments are supported.
- ☐ Bank transfer payments are supported.
- ☐ POS terminal payments are supported.
- ☐ USSD payments are supported where applicable.
- ☐ Credit sales are supported.
- ☐ Split payments are supported or explicitly deferred.

- ☐ Discount permissions exist.
- ☐ Sale cancellation permissions exist.
- ☐ Return permissions exist.
- ☐ Customer requirement for credit sales is configurable.
- ☐ Sale numbering is configured.
- ☐ Payment references can be recorded.
- ☐ Sales settings are audited.

Receipt settings

- ☐ Receipt business name.
- ☐ Receipt logo.
- ☐ Receipt phone.
- ☐ Receipt address.
- ☐ Receipt footer.
- ☐ Return policy.
- ☐ Tax display.
- ☐ Receipt numbering.
- ☐ Payment method.
- ☐ Transaction reference.
- ☐ Amount paid.
- ☐ Amount due.
- ☐ Receipt print support.
- ☐ Receipt download support.
- ☐ Receipt sharing support.
- ☐ Receipt settings are branch-aware.
- ☐ Historical receipts remain unchanged after settings updates.

5. Inventory Roles

Roles

- ☐ Inventory Owner role exists.
- ☐ Inventory Manager role exists.
- ☐ Sales Attendant role exists.
- ☐ Stock Manager role exists.
- ☐ Accountant role exists.

- ☐ Inventory Viewer role exists.

Permissions

- ☐ inventory.view.
- ☐ inventory.manage_products.
- ☐ inventory.create_products.
- ☐ inventory.edit_products.
- ☐ inventory.archive_products.
- ☐ inventory.restore_products.
- ☐ inventory.view_stock.
- ☐ inventory.receive_stock.
- ☐ inventory.adjust_stock.
- ☐ inventory.start_stock_count.
- ☐ inventory.approve_stock_adjustment.
- ☐ inventory.record_sales.
- ☐ inventory.cancel_sales.
- ☐ inventory.process_returns.
- ☐ inventory.view_cost.
- ☐ inventory.view_profit.
- ☐ inventory.view_reports.
- ☐ inventory.export_data.
- ☐ inventory.manage_members.
- ☐ inventory.manage_branch_access.
- ☐ inventory.manage_settings.
- ☐ inventory.manage_billing, if authorized.

Permission rules

- ☐ Permissions are checked server-side.
- ☐ Frontend visibility is not the security boundary.
- ☐ Inventory role is separate from organization role.
- ☐ Branch permissions are separate from Inventory permissions.
- ☐ User may have different Inventory roles in different organizations.
- ☐ User may have different branch assignments.
- ☐ Role changes are audited.
- ☐ Permission changes are audited.

- ☐ Owner cannot lose the only owner role without transfer.
- ☐ Staff cannot grant themselves permissions.
- ☐ Staff cannot access cost or profit without permission.
- ☐ Sales attendants cannot silently edit posted sales.
- ☐ Viewers cannot perform mutations.

6. Branch Foundation

Branch record

- ☐ Branch belongs to an organization.
- ☐ Branch belongs to Inventory for the MVP.
- ☐ Branch has a unique ID.
- ☐ Branch has a name.
- ☐ Branch has a code.
- ☐ Branch has an optional description.
- ☐ Branch has an address.
- ☐ Branch has a phone.
- ☐ Branch has an email.
- ☐ Branch has a manager, if required.
- ☐ Branch has a timezone, if different.
- ☐ Branch has a currency, if supported.
- ☐ Branch has a primary indicator.
- ☐ Branch has a status.
- ☐ Branch has created-by information.
- ☐ Branch has creation and update timestamps.
- ☐ Branch records include workspaceId.
- ☐ Branch records include productKey.
- ☐ Branch queries use workspace-scoped indexes.
- ☐ Branch queries use application-scoped indexes.

Branch statuses

- ☐ creating.
- ☐ active.
- ☐ setup_incomplete.
- ☐ suspended.

- ☐ archived.
- ☐ deleting, if deletion is later supported.
- ☐ deleted, if deletion is later supported.

Branch lifecycle rules

- ☐ First Inventory activation creates a primary branch.
- ☐ Primary branch can be renamed.
- ☐ Branch code is unique within the organization.
- ☐ Branch limit is checked server-side.
- ☐ Branch creation is idempotent.
- ☐ At least one active branch remains where Inventory requires it.
- ☐ Primary branch cannot be archived without a replacement.
- ☐ Suspended branches block normal operations.
- ☐ Archived branches preserve history.
- ☐ Branch deletion is disabled or restricted in the MVP.
- ☐ Branch status changes are audited.
- ☐ Branch status changes notify affected managers.
- ☐ Branches cannot be moved between organizations.
- ☐ Branches cannot be accessed through another organization's context.

7. Branch Creation

Branch creation form

- ☐ Branch name.
- ☐ Branch code.
- ☐ Description.
- ☐ Address.
- ☐ State.
- ☐ City.
- ☐ Local government area.
- ☐ Phone.
- ☐ Email.
- ☐ Manager.
- ☐ Opening hours.
- ☐ Logo, if supported.

- [] Receipt footer.
- [] Operational settings.

Branch creation flow

- [] User opens Inventory branch settings.
- [] User selects Add Branch.
- [] Backend authenticates user.
- [] Backend resolves organization.
- [] Backend verifies Inventory membership.
- [] Backend verifies branch-management permission.
- [] Backend checks organization status.
- [] Backend checks Inventory status.
- [] Backend checks billing entitlement.
- [] Backend checks branch limit.
- [] Backend validates branch fields.
- [] Backend validates unique branch code.
- [] Branch record is created.
- [] Branch settings are created.
- [] Optional branch manager is assigned.
- [] Audit event is created.
- [] Notification is sent.
- [] User is redirected to branch settings.
- [] Retry does not create duplicate branches.

Branch editing

- [] Authorized user can edit branch name.
- [] Authorized user can edit branch code where permitted.
- [] Authorized user can edit branch address.
- [] Authorized user can edit branch phone.
- [] Authorized user can edit branch email.
- [] Authorized user can edit opening hours.
- [] Authorized user can change manager.
- [] Authorized user can update branch operational settings.
- [] Edits are validated.
- [] Edits are audited.

- ☐ Historical records are not rewritten.
- ☐ Branch contact information is not treated as personal user information.
- ☐ Branch phone verification is optional unless used for SMS or security.

8. Branch Settings

General

- ☐ Branch profile page exists.
- ☐ Organization name is displayed.
- ☐ Inventory application is displayed.
- ☐ Branch name is displayed.
- ☐ Branch code is displayed.
- ☐ Branch status is displayed.
- ☐ Primary branch indicator is displayed.
- ☐ Created and updated dates are displayed.

Contact and location

- ☐ Branch address can be edited.
- ☐ Branch phone can be edited.
- ☐ Branch email can be edited.
- ☐ Branch phone is optional.
- ☐ Branch phone is normally unverified.
- ☐ Branch phone verification is triggered only for SMS or security use.
- ☐ Branch contact changes are audited.
- ☐ Branch contact information can appear on receipts.
- ☐ Branch contact information can appear in customer-facing views.
- ☐ Branch contact information is separate from organization contact information.

Opening hours

- ☐ Opening hours can be configured.
- ☐ Closing hours can be configured.
- ☐ Business days can be configured.
- ☐ Holiday closures can be supported or deferred.
- ☐ Timezone is respected.
- ☐ Opening hours do not automatically block emergency administrative access.

- ☐ Opening-hours changes are audited.

Branch operational settings

- ☐ Branch negative-stock policy can be configured.
- ☐ Branch low-stock threshold can be configured.
- ☐ Branch receipt footer can be configured.
- ☐ Branch payment methods can be configured.
- ☐ Branch stock-transfer permissions can be configured.
- ☐ Branch adjustment approval policy can be configured.
- ☐ Branch settings do not override higher-level entitlement limits.
- ☐ Branch settings do not override organization currency without explicit support.
- ☐ Branch settings changes are audited.

9. Branch Team Management

Membership levels

- ☐ Organization membership exists.
- ☐ Inventory membership exists.
- ☐ Branch membership exists.
- ☐ Branch role exists.
- ☐ Branch permission scope exists.
- ☐ Member can access one branch.
- ☐ Member can access multiple branches.
- ☐ Member can access all branches only when explicitly permitted.
- ☐ Member can belong to organization without Inventory access.
- ☐ Inventory member can lack access to a branch.

Team list

- ☐ Staff name.
- ☐ Staff email.
- ☐ Profile image.
- ☐ Organization role.
- ☐ Inventory role.
- ☐ Branch role.
- ☐ Branch assignments.

- ☐ Membership status.
- ☐ Joined date.
- ☐ Last active date.
- ☐ Invitation status.
- ☐ Actions based on permission.

Team actions

- ☐ Invite staff.
- ☐ Select Inventory role.
- ☐ Select branch access.
- ☐ Grant branch access.
- ☐ Revoke branch access.
- ☐ Change branch role.
- ☐ Transfer staff between branches.
- ☐ Preserve role during transfer where selected.
- ☐ Change role during transfer where selected.
- ☐ Suspend branch access.
- ☐ Restore branch access.
- ☐ Remove branch membership.
- ☐ Remove Inventory membership.
- ☐ Remove organization membership where authorized.
- ☐ Revoke relevant sessions.
- ☐ Notify affected user.
- ☐ Audit every access change.

Invitation rules

- ☐ Invitation token is cryptographically secure.
- ☐ Only token hash is stored.
- ☐ Token is single-use.
- ☐ Token expires.
- ☐ Invitation displays organization.
- ☐ Invitation displays Inventory.
- ☐ Invitation displays role.
- ☐ Invitation displays branch access.
- ☐ Invitation displays expiry.

- ☐ Wrong users cannot accept.
- ☐ Invitation does not grant access before acceptance.
- ☐ Duplicate pending invitations are prevented.
- ☐ Sender can revoke invitation.
- ☐ Invitee can decline invitation.
- ☐ Acceptance is audited.
- ☐ Invitation delivery is recorded.

10. Inventory Data Model

Product catalogue

- ☐ Product ID.
- ☐ Organization ID.
- ☐ Branch applicability.
- ☐ Product name.
- ☐ SKU.
- ☐ Barcode.
- ☐ Category.
- ☐ Brand.
- ☐ Description.
- ☐ Image.
- ☐ Base unit.
- ☐ Purchase cost.
- ☐ Retail price.
- ☐ Wholesale price.
- ☐ Reorder level.
- ☐ Preferred supplier.
- ☐ Tax status.
- ☐ Active status.
- ☐ Created by.
- ☐ Created date.
- ☐ Updated date.
- ☐ Archived status.
- ☐ Historical references preserved.

Stock

- ☐ Stock balance.
- ☐ Organization ID.
- ☐ Branch ID.
- ☐ Product ID.
- ☐ Quantity.
- ☐ Reserved quantity, if needed.
- ☐ Last movement date.
- ☐ Stock status.
- ☐ Low-stock status.
- ☐ Negative-stock status.

Stock movements

- ☐ Movement ID.
- ☐ Organization ID.
- ☐ Branch ID.
- ☐ Product ID.
- ☐ Quantity.
- ☐ Unit.
- ☐ Movement type.
- ☐ Reason.
- ☐ Related transaction.
- ☐ User ID.
- ☐ Device ID where offline.
- ☐ Timestamp.
- ☐ Sync status.
- ☐ Reversal reference.
- ☐ Audit reference.

Movement types

- ☐ Opening stock.
- ☐ Purchase received.
- ☐ Sale.
- ☐ Customer return.
- ☐ Supplier return.

- ☐ Transfer out.
- ☐ Transfer in.
- ☐ Damage.
- ☐ Expiry.
- ☐ Loss.
- ☐ Theft.
- ☐ Manual increase.
- ☐ Manual decrease.
- ☐ Stock-count correction.

Sales

- ☐ Sale ID.
- ☐ Organization ID.
- ☐ Branch ID.
- ☐ User ID.
- ☐ Customer ID.
- ☐ Sale lines.
- ☐ Payment information.
- ☐ Total.
- ☐ Discount.
- ☐ Tax.
- ☐ Amount paid.
- ☐ Amount due.
- ☐ Payment method.
- ☐ Transaction reference.
- ☐ Status.
- ☐ Receipt number.
- ☐ Created timestamp.
- ☐ Sync status.
- ☐ Idempotency key.
- ☐ Reversal or return reference.

Purchases

- ☐ Purchase ID.
- ☐ Organization ID.
- ☐ Branch ID.
- ☐ Supplier ID.
- ☐ Purchase lines.
- ☐ Expected quantity.
- ☐ Received quantity.
- ☐ Cost.
- ☐ Payment amount.
- ☐ Supplier balance.
- ☐ Purchase status.
- ☐ Receipt reference.
- ☐ Created timestamp.
- ☐ Confirmed timestamp.

11. Inventory Data Integrity

- ☐ Every tenant-owned record includes organization ID.
- ☐ Every branch-owned record includes branch ID.
- ☐ Every Inventory record includes application context where required.
- ☐ Stock cannot be changed directly.
- ☐ Every stock change creates a stock movement.
- ☐ Posted sales cannot be silently edited.
- ☐ Posted purchases cannot be silently edited.
- ☐ Corrections use reversals or adjustments.
- ☐ Sale confirmation and stock deduction are atomic.
- ☐ Confirmed purchase increases stock exactly once.
- ☐ Draft purchase does not change stock.
- ☐ Duplicate sync requests do not create duplicate sales.
- ☐ Duplicate requests use idempotency keys.
- ☐ Archived products remain available in history.
- ☐ Returns reference original sales where possible.
- ☐ Customer balances update accurately.
- ☐ Supplier balances update accurately.

- ☐ Historical receipts preserve original values.
- ☐ Large stock discrepancies require approval where configured.
- ☐ All sensitive operations create audit records.

The PRD explicitly requires atomic stock operations, idempotency, preserved history, and no silent stock changes.^[1]

12. Offline and Synchronization

- ☐ Offline product cache exists or is explicitly deferred.
- ☐ Offline stock snapshot exists or is explicitly deferred.
- ☐ Offline sales can be queued where supported.
- ☐ Local transaction ID is stored.
- ☐ Server transaction ID is stored after synchronization.
- ☐ Device ID is stored.
- ☐ Organization ID is stored.
- ☐ Branch ID is stored.
- ☐ User ID is stored.
- ☐ Created timestamp is stored.
- ☐ Sync status is stored.
- ☐ Retry count is stored.
- ☐ Error message is stored.
- ☐ Idempotency key is stored.
- ☐ Automatic retry exists.
- ☐ Exponential backoff exists.
- ☐ Failed records remain visible.
- ☐ Manual retry exists.
- ☐ Conflicts are visible.
- ☐ Conflicts are not silently overwritten.
- ☐ Duplicate offline sales are prevented.
- ☐ Offline data retention is defined.
- ☐ Unsynchronized records are clearly identified in reports.
- ☐ Local data is cleared only according to retention policy.

13. Inventory Dashboard

- ☐ Current organization is displayed.
- ☐ Current Inventory application is displayed.
- ☐ Current branch is displayed.
- ☐ Branch switcher exists.
- ☐ User role is displayed where useful.
- ☐ Today's sales are displayed.
- ☐ Transaction count is displayed.
- ☐ Cash received is displayed.
- ☐ Transfer received is displayed.
- ☐ POS received is displayed.
- ☐ Credit sales are displayed.
- ☐ Estimated gross profit is displayed.
- ☐ Low-stock count is displayed.
- ☐ Customer debt is displayed.
- ☐ Supplier debt is displayed.
- ☐ Pending sync count is displayed.
- ☐ Recent adjustments are displayed.
- ☐ Reports are branch-filtered.
- ☐ Reports exclude cancelled transactions.
- ☐ Reports reflect returns.
- ☐ Reports identify unsynchronized data.
- ☐ Dashboard permissions are enforced.
- ☐ Dashboard does not leak another branch's data.

14. Inventory Reports

- ☐ Sales by date.
- ☐ Sales by product.
- ☐ Sales by category.
- ☐ Sales by staff.
- ☐ Sales by payment method.
- ☐ Purchases.
- ☐ Stock on hand.
- ☐ Stock valuation.

- ☐ Low stock.
- ☐ Stock movements.
- ☐ Customer debt.
- ☐ Supplier balance.
- ☐ Returns.
- ☐ Expenses.
- ☐ Estimated profit.
- ☐ Stock adjustments.
- ☐ Branch filter.
- ☐ Product filter.
- ☐ Category filter.
- ☐ Customer filter.
- ☐ Supplier filter.
- ☐ Staff filter.
- ☐ Payment-method filter.
- ☐ Status filter.
- ☐ Date range filter.
- ☐ Report generation timestamp.
- ☐ Permission-controlled exports.
- ☐ Export activity is audited.
- ☐ Print-friendly output exists.
- ☐ Report calculations are documented.
- ☐ Costing method is documented.
- ☐ Discount treatment is documented.
- ☐ Return treatment is documented.
- ☐ Tax treatment is documented.
- ☐ Expense treatment is documented.
- ☐ Unsynchronized data treatment is documented.

15. Notifications

Inventory notifications

- ☐ Inventory activated.
- ☐ Inventory onboarding incomplete.
- ☐ Product import completed.
- ☐ Product import failed.
- ☐ Low stock.
- ☐ Stock adjustment.
- ☐ Sale completed.
- ☐ Purchase received.
- ☐ Return processed.
- ☐ Sync failure.
- ☐ Branch transfer.
- ☐ Staff invitation.
- ☐ Staff role changed.
- ☐ Branch access suspended.
- ☐ Branch access removed.
- ☐ Large stock discrepancy.
- ☐ Payment failure affecting Inventory access.
- ☐ Inventory suspension.
- ☐ Inventory restoration.

Notification controls

- ☐ In-app notifications exist.
- ☐ Email notifications exist.
- ☐ SMS is optional and feature-dependent.
- ☐ WhatsApp is optional and provider-dependent.
- ☐ Security notifications remain enabled.
- ☐ Users can control non-critical alerts.
- ☐ Owners receive important operational alerts.
- ☐ Managers receive branch-relevant alerts.
- ☐ Sales attendants do not receive unnecessary billing alerts.
- ☐ Notification preferences are scoped correctly.
- ☐ Notification delivery status is tracked.
- ☐ Failed notification delivery is visible to support.

16. Superadmin Inventory Access

Superadmin can view

- ☐ Inventory application registry record.
- ☐ Inventory status.
- ☐ Inventory activation date.
- ☐ Inventory entitlement.
- ☐ Inventory onboarding state.
- ☐ Organization using Inventory.
- ☐ Inventory application members.
- ☐ Inventory roles.
- ☐ Inventory permissions.
- ☐ Branch list.
- ☐ Branch status.
- ☐ Branch managers.
- ☐ Branch member counts.
- ☐ Branch product counts.
- ☐ Branch transaction counts.
- ☐ Branch stock summaries.
- ☐ Inventory usage.
- ☐ Product import history.
- ☐ Failed imports.
- ☐ Sync failures.
- ☐ Inventory audit logs.
- ☐ Branch audit logs.
- ☐ Application configuration.
- ☐ Application billing state.
- ☐ Entitlement mismatch alerts.

Superadmin may perform

- ☐ View Inventory status.
- ☐ Suspend Inventory.
- ☐ Restore Inventory.
- ☐ Retry failed application provisioning.
- ☐ Recalculate Inventory entitlements.

- ☐ Suspend a branch.
- ☐ Restore a branch.
- ☐ Archive a branch.
- ☐ Restore an archived branch.
- ☐ Suspend branch access.
- ☐ Restore branch access.
- ☐ Revoke organization sessions.
- ☐ Add support notes.
- ☐ Review failed synchronization.
- ☐ Review failed imports.
- ☐ Export permitted summaries.
- ☐ Extend trial with a reason.
- ☐ Review usage-limit disputes.

Superadmin restrictions

- ☐ Cannot directly edit stock balances.
- ☐ Cannot directly edit posted sales.
- ☐ Cannot directly edit posted purchases.
- ☐ Cannot silently change Inventory roles.
- ☐ Cannot silently grant branch access.
- ☐ Cannot delete historical stock movements.
- ☐ Cannot delete audit logs.
- ☐ Cannot view user passwords.
- ☐ Cannot view session tokens.
- ☐ Cannot view TOTP secrets.
- ☐ Cannot impersonate users in the MVP.
- ☐ Cannot change personal user data from Inventory administration.
- ☐ Cannot bypass audit requirements.
- ☐ Cannot bypass organization billing without a controlled override.

17. Inventory Audit Logs

Application events

- ☐ Inventory activation started.
- ☐ Inventory activated.
- ☐ Inventory suspended.
- ☐ Inventory restored.
- ☐ Inventory deactivated.
- ☐ Inventory onboarding started.
- ☐ Inventory onboarding step completed.
- ☐ Inventory onboarding step skipped.
- ☐ Inventory onboarding completed.
- ☐ Inventory provisioning failed.
- ☐ Inventory provisioning retried.
- ☐ Inventory settings updated.
- ☐ Inventory role changed.
- ☐ Inventory permission changed.
- ☐ Inventory export requested.

Product events

- ☐ Product created.
- ☐ Product updated.
- ☐ Product archived.
- ☐ Product restored.
- ☐ Product imported.
- ☐ Product import failed.
- ☐ Product exported.
- ☐ Price changed.
- ☐ Barcode changed.
- ☐ SKU changed.

Stock events

- ☐ Stock received.
- ☐ Stock sold.
- ☐ Stock adjusted.
- ☐ Stock counted.
- ☐ Stock transferred.

- ☐ Stock damaged.
- ☐ Stock expired.
- ☐ Stock lost.
- ☐ Stock returned.
- ☐ Stock discrepancy approved.
- ☐ Stock movement reversed.

Sales and purchase events

- ☐ Sale created.
- ☐ Sale confirmed.
- ☐ Sale cancelled.
- ☐ Sale returned.
- ☐ Payment recorded.
- ☐ Purchase created.
- ☐ Purchase confirmed.
- ☐ Purchase cancelled.
- ☐ Supplier payment recorded.
- ☐ Customer payment recorded.

Branch events

- ☐ Branch created.
- ☐ Branch updated.
- ☐ Branch set as primary.
- ☐ Branch suspended.
- ☐ Branch restored.
- ☐ Branch archived.
- ☐ Branch member added.
- ☐ Branch member removed.
- ☐ Branch role changed.
- ☐ Branch access granted.
- ☐ Branch access revoked.

Audit rules

- ☐ Audit logs are append-only.
- ☐ Every stock-changing operation creates an audit event.
- ☐ Every stock-changing operation creates a stock movement.
- ☐ Audit logs include actor.
- ☐ Audit logs include organization.
- ☐ Audit logs include application.
- ☐ Audit logs include branch.
- ☐ Audit logs include target.
- ☐ Audit logs include timestamp.
- ☐ Audit logs include request ID.
- ☐ Audit logs include safe metadata.
- ☐ Audit logs exclude passwords.
- ☐ Audit logs exclude tokens.
- ☐ Audit logs exclude payment secrets.
- ☐ Audit logs exclude TOTP secrets.
- ☐ Audit logs exclude unnecessary personal data.

18. Branch Notifications and Audit

- ☐ Branch created notification.
- ☐ Branch updated notification.
- ☐ Branch archived notification.
- ☐ Branch restored notification.
- ☐ Branch suspended notification.
- ☐ Branch manager changed notification.
- ☐ Branch member added notification.
- ☐ Branch member removed notification.
- ☐ Branch access suspended notification.
- ☐ Branch access restored notification.
- ☐ Branch transfer notification.
- ☐ Branch-specific low-stock notification.
- ☐ Branch-specific import failure notification.
- ☐ Branch-specific sync failure notification.
- ☐ Branch events appear in organization audit logs.

- [] Branch events appear in Inventory audit logs.
- [] Branch members see only relevant notifications.
- [] Owners and managers receive important branch notifications.

19. API Checklist

Application access

- [] GET /v1/workspaces/:workspaceId/applications.
- [] GET /v1/workspaces/:workspaceId/applications/inventory.
- [] POST /v1/workspaces/:workspaceId/applications/inventory/activate.
- [] POST /v1/workspaces/:workspaceId/applications/inventory/suspend.
- [] POST /v1/workspaces/:workspaceId/applications/inventory/restore.

Inventory settings

- [] GET /v1/workspaces/:workspaceId/inventory/settings.
- [] PATCH /v1/workspaces/:workspaceId/inventory/settings.
- [] GET /v1/workspaces/:workspaceId/inventory/roles.
- [] PATCH /v1/workspaces/:workspaceId/inventory/roles.
- [] GET /v1/workspaces/:workspaceId/inventory/audit.

Branches

- [] GET /v1/workspaces/:workspaceId/branches.
- [] POST /v1/workspaces/:workspaceId/branches.
- [] GET /v1/workspaces/:workspaceId/branches/:branchId.
- [] PATCH /v1/workspaces/:workspaceId/branches/:branchId.
- [] POST /v1/workspaces/:workspaceId/branches/:branchId/set-primary.
- [] POST /v1/workspaces/:workspaceId/branches/:branchId/suspend.
- [] POST /v1/workspaces/:workspaceId/branches/:branchId/restore.
- [] POST /v1/workspaces/:workspaceId/branches/:branchId/archive.
- [] GET /v1/workspaces/:workspaceId/branches/:branchId/audit.

Branch teams

- [] GET /v1/workspaces/:workspaceId/branches/:branchId/members.
- [] POST /v1/workspaces/:workspaceId/branches/:branchId/members.
- [] PATCH /v1/workspaces/:workspaceId/branches/:branchId/members/:membershipId.

- `[] DELETE /v1/workspaces/:workspaceId/branches/:branchId/members/:membershipId.`
- `[] POST /v1/workspaces/:workspaceId/branches/:branchId/members/:membershipId/suspend.`
- `[] POST /v1/workspaces/:workspaceId/branches/:branchId/members/:membershipId/restore.`

Inventory operations

- `[]` Product endpoints are organization-scoped.
- `[]` Stock endpoints are branch-scoped.
- `[]` Sales endpoints are branch-scoped.
- `[]` Purchase endpoints are branch-scoped.
- `[]` Customer endpoints are organization-scoped with branch filters.
- `[]` Supplier endpoints are organization-scoped with branch filters.
- `[]` Report endpoints enforce branch access.
- `[]` Export endpoints enforce permissions.
- `[]` Offline sync endpoints require idempotency.
- `[]` Every mutation validates organization and branch ownership.

20. Security and Access Rules

- `[]` Authenticate every protected request.
- `[]` Resolve workspace context server-side.
- `[]` Verify active organization membership.
- `[]` Verify Inventory membership.
- `[]` Verify branch membership.
- `[]` Verify role.
- `[]` Verify permission.
- `[]` Verify organization status.
- `[]` Verify Inventory status.
- `[]` Verify branch status.
- `[]` Verify billing entitlement.
- `[]` Validate resource ownership.
- `[]` Reject forged organization IDs.
- `[]` Reject forged application keys.
- `[]` Reject forged branch IDs.
- `[]` Reject cross-tenant object access.

- ☐ Revoke access after membership removal.
- ☐ Revoke or refresh sessions after sensitive membership changes.
- ☐ Rate-limit invitations.
- ☐ Rate-limit exports.
- ☐ Rate-limit high-risk operations.
- ☐ Do not expose hidden Inventory data through broad queries.
- ☐ Do not rely on frontend branch filters for security.
- ☐ Do not rely on frontend roles for authorization.
- ☐ Log suspicious cross-organization attempts.
- ☐ Protect uploaded product and import files.
- ☐ Validate CSV files.
- ☐ Reject malicious file uploads.
- ☐ Apply file-retention rules.

The workspace plan requires authentication, workspace resolution, membership validation, product access validation, permission checks, workspace-scoped queries, and resource ownership confirmation for every protected operation.^[2]

21. Privacy and Data Protection

- ☐ Organization data is separated from personal account data.
- ☐ Personal phone numbers are not exposed by default.
- ☐ Personal email is not editable by workspace administrators.
- ☐ Personal security sessions are not exposed to branch managers.
- ☐ Staff profiles show only necessary information.
- ☐ Business phone is separate from personal phone.
- ☐ Branch phone is separate from personal phone.
- ☐ Exported Inventory data is organization-scoped.
- ☐ Exported Inventory data is permission-controlled.
- ☐ Export activity is audited.
- ☐ Product images use protected storage where necessary.
- ☐ Uploaded CSV files are deleted after processing or retention period.
- ☐ Personal data in audit logs is minimized.
- ☐ Historical business records remain accurate.
- ☐ Deleted users do not automatically delete business records.
- ☐ Staff-created records retain appropriate attribution.
- ☐ Anonymization policy exists for deleted accounts.

- ☐ Retention rules exist for sales, purchases, stock, payments, and audit logs.
- ☐ Privacy notices explain Inventory data use.
- ☐ No unsupported tax or regulatory compliance claim is displayed.

22. Do's

- ☐ Do scope Inventory data to the organization.
- ☐ Do scope stock and operational transactions to the branch.
- ☐ Do make Inventory access separate from organization membership.
- ☐ Do create a primary branch during Inventory setup.
- ☐ Do enforce branch limits server-side.
- ☐ Do use explicit Inventory roles.
- ☐ Do use explicit branch permissions.
- ☐ Do create stock movements for every stock change.
- ☐ Do preserve posted transaction history.
- ☐ Do use reversals and adjustments instead of silent edits.
- ☐ Do use idempotency keys for sales, imports, payments, and sync.
- ☐ Do make onboarding resumable.
- ☐ Do keep future applications hidden.
- ☐ Do show the active organization and branch in the interface.
- ☐ Do make reports branch-aware.
- ☐ Do audit staff access changes.
- ☐ Do audit administrative changes.
- ☐ Do notify users about important access changes.
- ☐ Do preserve historical data when archiving branches.
- ☐ Do support offline operation carefully where implemented.
- ☐ Do show synchronization failures instead of hiding them.
- ☐ Do validate CSV uploads.
- ☐ Do require approval for large stock discrepancies where configured.
- ☐ Do protect sensitive product cost and profit information.
- ☐ Do allow superadmin to inspect safely without editing business data directly.

23. Don'ts

- ☐ Do not treat organization membership as Inventory access.
- ☐ Do not treat Inventory membership as access to every branch.
- ☐ Do not trust client-supplied organization IDs.
- ☐ Do not trust client-supplied branch IDs.
- ☐ Do not trust client-supplied roles.
- ☐ Do not allow users to access another organization's branch.
- ☐ Do not allow direct stock balance editing.
- ☐ Do not silently edit posted sales.
- ☐ Do not silently edit posted purchases.
- ☐ Do not delete historical stock movements.
- ☐ Do not delete archived products from transaction history.
- ☐ Do not create branches without entitlement checks.
- ☐ Do not let suspended branches continue normal operations.
- ☐ Do not allow the only primary branch to be archived.
- ☐ Do not let future applications appear in Inventory.
- ☐ Do not allow future application permissions in the MVP.
- ☐ Do not expose another branch's stock in reports.
- ☐ Do not create duplicate sales during offline synchronization.
- ☐ Do not silently overwrite sync conflicts.
- ☐ Do not store raw invitation tokens.
- ☐ Do not store passwords or authentication secrets in Inventory records.
- ☐ Do not expose private user security information to branch managers.
- ☐ Do not let superadmin silently modify stock or sales.
- ☐ Do not delete business data when a staff member leaves.
- ☐ Do not block all users because one staff member is removed.
- ☐ Do not claim full accounting, tax, or e-invoicing compliance unless implemented and verified.

24. Testing Checklist

Application tests

- ☐ Inventory appears in the launcher.
- ☐ Future applications remain hidden.
- ☐ Inventory activation works.
- ☐ Duplicate activation is safe.
- ☐ Suspended Inventory blocks restricted access.
- ☐ Restored Inventory becomes available.
- ☐ Users without Inventory membership are rejected.
- ☐ Users with inactive organization membership are rejected.
- ☐ Direct URL manipulation fails.
- ☐ Cross-organization Inventory access fails.

Onboarding tests

- ☐ Primary branch is created.
- ☐ Inventory onboarding can resume.
- ☐ Required steps cannot be skipped.
- ☐ Optional steps can be skipped.
- ☐ Failed import can be retried.
- ☐ Failed onboarding does not create duplicates.
- ☐ Completion state is persisted.
- ☐ Onboarding events are audited.

Branch tests

- ☐ User can create a branch within plan limit.
- ☐ User cannot exceed branch limit.
- ☐ Duplicate branch code is rejected.
- ☐ Branch belongs to the correct organization.
- ☐ Branch belongs to Inventory.
- ☐ Primary branch cannot be improperly archived.
- ☐ Branch suspension works.
- ☐ Branch restoration works.
- ☐ Branch archive preserves history.
- ☐ Unauthorized branch access fails.
- ☐ Branch member assignment works.
- ☐ Branch transfer works.

- ☐ Removed branch member loses access.

Product and stock tests

- ☐ Product creation works.
- ☐ Duplicate SKU is handled.
- ☐ Duplicate barcode is handled.
- ☐ Product archive works.
- ☐ Archived product remains in history.
- ☐ Stock receipt creates movement.
- ☐ Sale creates movement.
- ☐ Adjustment creates movement.
- ☐ Return creates movement.
- ☐ Draft purchase does not change stock.
- ☐ Sale and stock deduction are atomic.
- ☐ Posted records are not silently edited.
- ☐ Large discrepancy approval works.

Offline tests

- ☐ Offline sale is queued.
- ☐ Offline sale has idempotency key.
- ☐ Offline sale synchronizes once.
- ☐ Duplicate sync is rejected.
- ☐ Failed sync remains visible.
- ☐ Retry works.
- ☐ Conflict is visible.
- ☐ Conflict is not silently overwritten.
- ☐ Branch ID is preserved during sync.
- ☐ Organization ID is preserved during sync.

Team tests

- ☐ Organization member can lack Inventory access.
- ☐ Inventory member can lack branch access.
- ☐ Branch access can be granted.
- ☐ Branch access can be revoked.
- ☐ Branch role changes correctly.

- ☐ Staff transfer preserves or changes role correctly.
- ☐ Removed staff retain historical attribution.
- ☐ Invitations expire.
- ☐ Revoked invitations cannot be accepted.
- ☐ Wrong users cannot accept invitations.

Superadmin tests

- ☐ Superadmin can view Inventory status.
- ☐ Superadmin can view branches.
- ☐ Superadmin can view team counts.
- ☐ Superadmin can view Inventory onboarding.
- ☐ Superadmin can view usage.
- ☐ Superadmin can view Inventory audit logs.
- ☐ Superadmin can suspend and restore Inventory.
- ☐ Superadmin can suspend and restore branches.
- ☐ Superadmin actions require confirmation.
- ☐ Superadmin actions are audited.
- ☐ Superadmin cannot directly edit stock.
- ☐ Superadmin cannot directly edit sales.
- ☐ Secrets are not displayed.

25. Launch Acceptance Criteria

Inventory application

- ☐ Inventory is the only available application.
- ☐ Inventory application access is organization-scoped.
- ☐ Inventory access is permission-controlled.
- ☐ Inventory onboarding works.
- ☐ Inventory onboarding is resumable.
- ☐ Primary branch is created automatically.
- ☐ Inventory settings exist.
- ☐ Inventory roles exist.
- ☐ Inventory permissions exist.
- ☐ Inventory audit logs exist.
- ☐ Inventory notifications exist.

- ☐ Inventory usage limits are enforced.
- ☐ Inventory dashboard shows the correct organization and branch.
- ☐ Inventory reports respect branch access.
- ☐ Inventory exports are controlled and audited.

Branches

- ☐ Branch creation works.
- ☐ Branch editing works.
- ☐ Branch settings work.
- ☐ Branch contact information works.
- ☐ Branch team management works.
- ☐ Branch roles and permissions work.
- ☐ Branch suspension works.
- ☐ Branch restoration works.
- ☐ Branch archive works.
- ☐ Branch limits are enforced.
- ☐ Primary branch protection works.
- ☐ Branch history is preserved.
- ☐ Branch audit logs exist.
- ☐ Branch notifications exist.
- ☐ Cross-branch access is blocked.

Data integrity

- ☐ Every tenant-owned record has organization context.
- ☐ Every branch-owned record has branch context.
- ☐ Every stock change has a stock movement.
- ☐ Posted transactions cannot be silently edited.
- ☐ Sale and stock deduction are atomic.
- ☐ Duplicate sync requests are safe.
- ☐ Duplicate imports are safe.
- ☐ Historical records are preserved.
- ☐ Audit logs are append-only.
- ☐ Sensitive information is excluded from logs.

Final definition of done

- ☐ A user can enter an organization.
- ☐ The user can activate Inventory.
- ☐ Inventory creates a primary branch.
- ☐ The user can complete onboarding.
- ☐ The user can create additional branches if entitled.
- ☐ The user can assign staff to selected branches.
- ☐ Staff receive only their assigned Inventory and branch access.
- ☐ Products, stock, sales, and purchases are correctly scoped.
- ☐ Reports show the correct branch data.
- ☐ Superadmin can inspect Inventory and branch state.
- ☐ Superadmin can perform safe administrative actions.
- ☐ All important changes are audited.
- ☐ No cross-organization or cross-branch data leakage exists.
- ☐ The system preserves business history.
- ☐ The system is ready for future applications without exposing them prematurely.

✱✱

1. docs/Orviohub full PRD.pdf
2. docs/Workspace Management Implementation Plan.pdf